

Labor Supply, Care Work, Fertility, And Household Allocation

Gender Study — Week 2

Teaching deck draft

Central question

- How do care responsibilities, fertility, and household allocation shape labor supply and earnings trajectories?
- This is labor economics because care work changes employment, hours, earnings, occupations, firms, schedules, and job quality.
- Week 2 discipline: match each empirical object to the margin it identifies.

Week 1 to Week 2 bridge

- Week 1: name the labor object before naming the mechanism.
- Week 2: move from person-level gaps to household and lifecycle allocation.
- A post-birth earnings gap can reflect care time, hours, employment, occupation, firm attachment, bargaining, policy, or norms.

Labor supply inside the household

Object	Labor-market margin
Market work	participation, hours, earnings, wage growth
Unpaid care	time constraints, availability for work, fatigue, schedule limits
Household production	substitution between parental time and market inputs
Fertility timing	interruptions, experience, promotion, firm continuity
Control over resources	labor supply, spending, care allocation, outside options

$$T_i = h_i + a_i + \ell_i$$

- h_i : market work.
- a_i : unpaid care or home production.
- ℓ_i : leisure.
- A birth shock raises care demand and changes the feasible allocation of time.

Household production

$$\max U(c_f, c_m, \ell_f, \ell_m, q) \quad \text{s.t.} \quad c_f + c_m + p_x x = w_f h_f + w_m h_m + y, \quad q = F(a_f, a_m, x)$$

- Care quality q is produced with parental time and market inputs.
- Childcare policy changes p_x , availability, reliability, or timing.
- The labor response can be participation, hours, earnings, occupation, firm choice, or job quality.

Why comparative advantage is not enough

- Wage differences can rationalize some specialization, but they do not explain all gendered care allocation.
- Persistent asymmetry may also reflect childcare constraints, bargaining, employer flexibility, and norms.
- The empirical task is to avoid labeling every post-birth outcome as preferences or as discrimination.

- Fertility changes care demand and lifecycle timing.
- Child penalties trace dynamic post-birth paths for mothers and fathers.
- Outcomes include employment, hours, earnings, occupation, firm attachment, promotion, and job quality.
- The child penalty is a labor-market object, not a mechanism by itself.

Child-penalty event-study object

$$Y_{it}^g = \sum_{k \neq -1} \beta_k^g \mathbf{1}\{t - B_i = k\} + \alpha_i + \gamma_t + \varepsilon_{it}$$

- B_i is first birth and k is event time.
- β_k^g traces the dynamic path relative to the pre-birth period.
- Best for timing and persistence; not automatically a causal fertility estimate.

Fertility effects are a different object

$$Y_i = \alpha + \tau K_i + X_i' \delta + u_i, \quad K_i \leftarrow Z_i$$

- K_i is fertility and Z_i is a fertility shifter such as IVF success.
- The target is a causal fertility effect for a specific margin.
- External validity depends on which families and fertility margins the shifter moves.

Bargaining and control over resources

$$\max \lambda U_f(c_f, \ell_f, q) + (1 - \lambda) U_m(c_m, \ell_m, q)$$

- Unitary models pool resources and provide a clean benchmark.
- Collective models allow resource control and outside options to move λ .
- A policy effect may be a price effect, income effect, or bargaining effect.

Childcare and policy constraints

- Childcare can relax a binding time constraint without changing wages.
- Single-parent and coupled households may respond on different margins.
- The relevant outcome may be employment, hours, earnings, retention, occupation, firm choice, or job quality.
- Policy evidence is strongest when it states the margin moved and the counterfactual.

Designs and what they identify

Setting	Object	Method	Main caution
Birth event study	child penalty path	event time / panel FE	descriptive by itself
IVF or infertility	causal fertility effect	IV / quasi-experiment	local margin
Childcare reform	constraint relaxation	RCT / DID	multiple channels
Resource-control shift	bargaining response	RF / structural	map shift to power
Time-use panel	care reallocation	descriptive / panel	weak on earnings dynamics

Margin discipline

- Extensive margin: does the worker participate?
- Intensive margin: how many hours and which schedule?
- Earnings: hours, wages, employment, and career progression combined.
- Job quality: flexibility, predictability, commute, promotion, authority, and firm attachment.
- Care time: unpaid work can move even when payroll outcomes do not.

Norms bridge to Week 5

- Domestic responsibility allocation partly reflects gender norms and traditions.
- Norms about caregiving, motherhood, breadwinning, and relative earnings can sustain specialization.
- Week 2 flags this mechanism; Week 5 studies norms, beliefs, identity, and institutions directly.

- Reproduce: synthetic child-penalty event-time summaries.
- Diagnose: what the event study shows, and what it cannot identify alone.
- Transfer: synthetic IVF-style fertility design and local causal fertility effects.
- Optional policy prompt: how childcare availability changes the same household allocation problem.

- Week 3: care responsibilities and fertility timing shape skills, aspirations, and occupational sorting.
- Week 5: norms and institutions help explain persistent domestic responsibility allocation.
- Later policy weeks ask which interventions move the actual labor margin: care time, employment, hours, earnings, or job quality.

Takeaways

- Care work is a production and allocation problem.
- Child penalties, fertility effects, bargaining shifts, and childcare policy effects identify different objects.
- The labor margins are multidimensional: participation, hours, earnings, occupations, firms, promotions, care time, and job quality.
- Week 2 is the household and lifecycle foundation for the rest of the course.